

Speech-to-Sign Project

Scope Reality Check

What is genuinely possible in 8 weeks — and what is not

Decision requested at the end of this presentation

Mothishwaran · Roll No. 41 · Section AIE-A

Speech Processing (22AIE450) · Amrita School of AI, Coimbatore

1

NOT POSSIBLE

2

RISKY STRETCH

3

WE CAN BUILD THIS

Why We Are Presenting This

We tested our own assumptions before writing any code

1

We wrote a plan

Our first plan assumed we could download ready-made sign animations and let the avatar play them.

2

We checked it

We searched for those animations, and tested whether AI video tools can generate signing.

3

Three assumptions failed

The parts of the plan that failed are the parts everything else depended on.

Rather than discover this in week six, we are bringing it to you in week zero.

This presentation separates the project into three tiers: what is not possible at all, what is possible but risky, and what we can commit to delivering. Each tier lists the exact tools and the week-by-week plan.

Our Original Plan, Checked Line by Line

Four of these six assumptions do not survive contact with reality

What our plan assumed	What we actually found	Status
Download sign animations from Mixamo	Mixamo has no sign language content at all	FAILS
AI avatar tools can perform the signs	They generate hand motion that is not real sign language	FAILS
Build a 50-word signing vocabulary	Each sign takes 1–3 hours to author by hand	FAILS
Fingerspell any word not in the dictionary	H-O-W-A-R-E-Y-O-U is unreadable in conversation	FAILS
Convert live speech to text in real time	Whisper on CPU handles this comfortably	HOLDS
Show the result in a web browser	Standard web technology, no obstacle	HOLDS

The four failures are all in the same place: producing the signing. The speech half of the project was never the problem.

The Project Split Into Three Tiers

Everything from here is organised under these headings



NOT POSSIBLE

6 items

No amount of effort or time in this semester makes these work. Some are unsolved research problems.



RISKY STRETCH

5 items

Technically achievable but likely to consume the whole semester. Attempted only after the core is finished.



WE COMMIT TO THIS

7 items

Fully within reach in eight weeks with free tools. This is what we promise to deliver.

We are not asking to reduce the workload. We are asking to spend it on the parts that will actually work.

Tier 1 — Not Possible

These are not effort problems. They are capability problems.



AI generating sign language on its own

No AI video tool has ever been trained on sign language. It produces motion that only looks meaningful.



Translating any English sentence correctly

Requires spatial grammar, classifiers and verb directionality — an open research problem worldwide.



Training our own sign generation model

Needs a large annotated corpus, GPU cluster and months of training. We have none of the three.



A 3D avatar with a large signing vocabulary

The animation data does not exist and must be authored by hand at 1–3 hours per sign.



Word-by-word live interpretation

Sign language reorders across a whole clause, so we must wait for the sentence to finish.



A usable product for Deaf people

Would require validation by native signers and the Deaf community. Far beyond a subject project.

Why AI Avatars Cannot Sign

We tested this ourselves — here is how anyone can verify it

What the AI tools actually do

- Trained on talking heads and general web video
- No annotated sign language corpus in training
- Output resembles signing, but carries no meaning

The five-minute test anyone can run

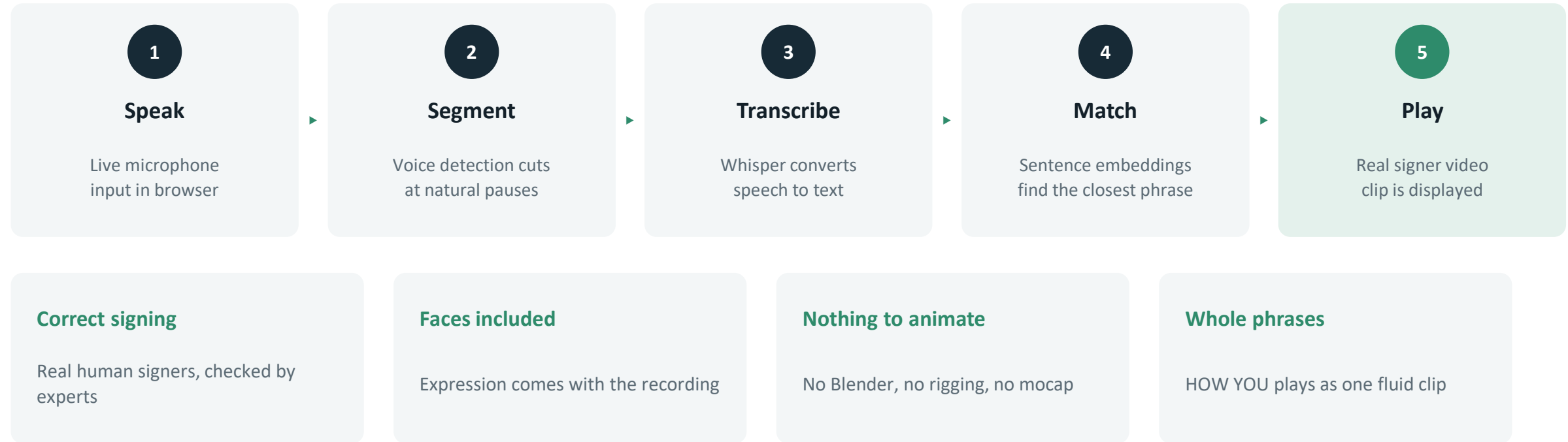
- ASL has no verb "to be" — no am, is or are
- "How are you" is signed HOW YOU — two signs
- **Count the movements. Three means it is not ASL.**

Even published research does not do this. Sign generation is split into two steps — mapping language to correct poses, then rendering those poses. AI video tools only ever attempt the second step, and the first step is the part nobody has solved.

The risk to us: AI-generated signing looks convincing to anyone who does not sign. Building our demo on it would mean submitting something that appears to work and does not.

Tier 3 — What We Can Definitely Build

Stop generating the signing. Retrieve it from real recorded signers.



Fingerspelling is used only for names and unknown words — which is exactly how real sign language uses it.

Tier 3 — Exact Tools We Will Use

Every item is free and runs on a normal laptop with no GPU

Stage	Tool	Specification
Capture	Web Audio API / MediaRecorder	16 kHz mono, in the browser
Segmenting	Silero VAD	Approx. 1 MB model, cuts on pauses
Recognition	faster-whisper (CTranslate2)	base.en, int8 quantised, CPU only
Backend	Python 3.11 + FastAPI	WebSocket connection, stays open
Matching	sentence-transformers	all-MiniLM-L6-v2 embeddings
Sign data	WLASL (ASL) or INCLUDE (ISL)	Free for academic use, real signers
Spelling	Alphabet clip set	Used only for names and unknowns
Frontend	React + HTML5 video	Preloads clips to avoid stutter
Evaluation	Python scripts	Logs latency and match accuracy

WLASL provides roughly 21,000 videos across 2,000 ASL words. INCLUDE provides 4,287 videos across 263 Indian Sign Language signs. Both are licensed for academic use.

Tier 3 — Eight-Week Delivery Plan

Each week ends with something we can show you working

Week	Work	Proof it is done
1	Collect and clean sign clips	40 clips downloaded, trimmed and named
2	Build the phrase bank	Typed word plays the correct sign clip
3	Speech recognition layer	Live microphone prints accurate text
4	Voice segmentation	System cuts cleanly at natural pauses
5	Semantic matching engine	Spoken sentence finds the right phrase
6	Fingerspelling fallback	Unknown names are spelled letter by letter
7	Interface and latency tuning	End-to-end lag measured and reported
8	Evaluation and documentation	Results, demo video, report and slides

There is no week in this plan that depends on something we have not already confirmed exists.

How We Will Prove It Works

Four measurable results, not just a demonstration

Latency	Under 2 s	Time from end of speech to first sign appearing
Word accuracy	Under 15 %	Word error rate of the speech recognition layer
Match accuracy	Over 85 %	Correct phrase chosen on our own test sentences
Coverage	Over 70 %	Utterances matched to a real sign rather than spelled

These are the numbers we will report, whether or not we reach the targets.

Tier 2 — Possible, But Risky

Attempted only after the Tier 3 system is complete and working

	Smooth blending between clips	Clips use different signers and backgrounds, so cuts look abrupt.	Medium risk
	A 3D avatar for a few signs	Five to eight signs hand-authored in Blender as a visual showcase.	High risk
	Facial expression on the 3D avatar	Eyebrow and head rules driven by blendshapes. Needs the avatar first.	High risk
	Word-level sentence assembly	Building unseen sentences from single signs. Grammar quality will be poor.	High risk
	Switching to Indian Sign Language	Fewer signs available, so we may need to record some ourselves.	Medium risk

Each of these could consume the entire semester on its own. None of them is a promise.

Tier 2 — Tools and Conditional Plan

This work starts only if Week 6 of the core plan finishes on time

Stretch goal	Tools required	When we would attempt it
Clip blending	OpenCV, rembg background removal	Week 7, alongside interface work
3D avatar showcase	Blender 4.x, ReadyPlayerMe, Three.js	Week 7 only if core is complete
Avatar facial layer	ARKit blendshapes, react-three-fiber	Week 8, demo of 2 phrases only
Word-level assembly	SpaCy en_core_web_sm, custom rules	Week 8, presented as future work
ISL migration	INCLUDE dataset, own recordings	Week 1 decision, not later

Our rule for this tier

No stretch goal is started until every Tier 3 milestone is signed off. If a stretch goal slips, we abandon it rather than let it damage the core.

One decision needed in Week 1

ASL gives us a far larger vocabulary. ISL is more relevant to India but has fewer recorded signs. We need your guidance on which to build.

What You Will See in Week 8

The exact demonstration we are committing to

- 1 We open the web page**
A microphone button and an empty video panel.
- 2 We speak a sentence**
"Hello, how are you? My name is Mothishwaran."
- 3 The system segments**
Voice detection splits it into three utterances.
- 4 Signing plays back**
HELLO, then HOW YOU, then MY NAME, then the name spelled out.
- 5 Numbers appear**
Measured latency and match confidence shown on screen.

Every sign shown is performed by a real human signer, not generated.

What We Will Never Claim

Stated openly in our report and presentation

- This is not a translation system. It matches speech to prepared signs.
- It does not handle sentences outside our prepared vocabulary.
- It does not reproduce sign language grammar, only individual signs and phrases.
- It is a demonstrator for a subject project, not an accessibility product.
- It does not replace a human interpreter and must never be presented as doing so.

Signing avatar projects are criticised in Deaf communities mainly for over-claiming. Naming our limits is part of doing this properly.

Do you still want us to proceed?

Three options, and we are ready to take any of them

A

Build Tier 3 only

The retrieval system, delivered fully and evaluated properly. Lowest risk, certain result.

Our recommendation

B

Tier 3, then attempt Tier 2

Same core, with stretch goals attempted in the final two weeks if time allows.

Higher risk

C

Change the project now

If the reduced scope is not acceptable for this subject, we would rather know in week one.

Still possible today

Whichever you choose, we will start this week.